

Multi-Agent System: Theories and Applications

Lecture 8: Edge Dynamics for Linear MASs

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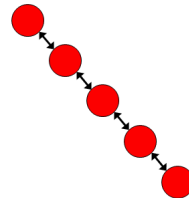
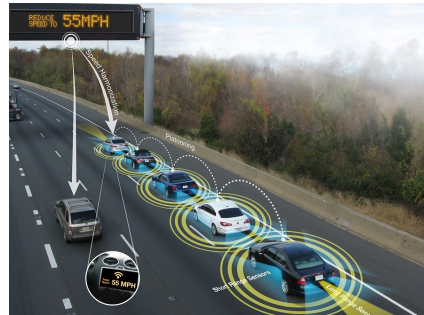
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Outline

- 1 Why edges important?
- 2 Edge dynamics
- 3 Consensus based on edge dynamics

Edges $\Leftrightarrow$ Inter-agent relation

- Dynamics of each agent is important, but inter-agent relation is also critical to achieve global goals of the whole MAS.
- Inter-agent relation is represented by edges in the MAS graph. For example, consensus of agents $\Leftrightarrow$ inter-agent states = 0.
- Hitherto, all MAS problems are studied with “node dynamics”, but this lecture gives another perspective of MASs from edges.



 agent (vehicle)

 inter-agent communication

Edge dynamics = mathematical equation for inter-agent relation

Main ideas

- Partition of MAS graph into a spanning tree $\mathcal{G}_t$ and the rest $\mathcal{G}_c$.
- Similarly, partitioning incidence matrix into $E(\mathcal{G}_t)$ and $E(\mathcal{G}_c)$.
- Only developed for consensus problem of integrator MAS.
- Edge Laplacian:

$$L_e = E^T E$$

where E is the incidence matrix.

Original edge dynamics (Zelazo & Mesbahi, 2011)

- Integrator agents, i.e. $A = 0, B = 1$.
- Closed-loop consensus dynamics:

$$\dot{x}(t) = -Lx(t) = -(EE^T)x(t)$$

- Edge dynamics for consensus of integrator MAS:

$$\dot{x}_t(t) = -L_e(\mathcal{G}_t)R(\mathcal{G})R(\mathcal{G})^T x_t(t)$$

$$z(t) = R(\mathcal{G})^T x_t(t)$$

For details, take a look at:

D. Zelazo and M. Mesbahi, "Edge agreement: Graph-theoretic performance bounds and passivity analysis", IEEE Transactions on Automatic Control, vol. 56, no. 3, pp. 544–555, 2011 (<https://ieeexplore.ieee.org/document/5504814>).

Another model for edge dynamics

Node Dynamics

- Generic linear dynamics of agents:

$$\dot{x}(t) = (I_N \otimes A)x(t) + (I_N \otimes B)u(t)$$

$$x(t) = [x_1(t)^T, \dots, x_N(t)^T]^T,$$

$$u(t) = [u_1(t)^T, \dots, u_N(t)^T]^T.$$

Edge Dynamics

- Define new vectors:

$$z(t) \triangleq (E^T \otimes I_n)x(t),$$

$$w(t) \triangleq (E^T \otimes I_m)u(t).$$

- Edge dynamics (developed by myself, 2017):

$$\dot{z}(t) = [(E^T L^\dagger E) \otimes A]z(t) + (I_M \otimes B)w(t)$$

where $L^\dagger$ is the pseudo-inverse of L ,
 M : number of edges.

Dinh Hoa Nguyen, "Reduced-Order Distributed Consensus Controller Design via Edge Dynamics", *IEEE Transactions on Automatic Control*, vol. 62, no. 1, pp. 475–480, 2017 (<https://ieeexplore.ieee.org/document/7452569>).

Algebraic properties of the edge Laplacian and related matrix

Eigenspectrum

Denote $\bar{L} = E^T L^\dagger E$.

- L_e has exactly $N - 1$ non-zero eigenvalues equal to positive eigenvalues of L , other eigenvalues of L_e if exist are 0.
- $\bar{L}$ has exactly $N - 1$ non-zero eigenvalues equal to 1, other eigenvalues of $\bar{L}$ if exist are 0.

Transformed edge dynamics

Orthogonalize $\bar{L} = U \Gamma U^T$, where $\bar{\Gamma} = \text{diag}\{0, I_{N-1}\}$. Denote $\tilde{z} \triangleq (U^T \otimes I_n)z$, $\tilde{w} \triangleq (U^T \otimes I_m)w$.

- Edge dynamics becomes:

$$\dot{\tilde{z}} = (\bar{\Gamma} \otimes A) \tilde{z} + (I_M \otimes B) \tilde{w}$$

equivalently

$$\dot{\tilde{z}}_1 = (I_{M-N+1} \otimes B) \tilde{w}_1$$

$$\dot{\tilde{z}}_2 = (I_{N-1} \otimes A) \tilde{z}_2 + (I_{N-1} \otimes B) \tilde{w}_2$$

which is called the transformed edge dynamics.

- $\tilde{z}_1$ and $\tilde{w}_1$ are always 0 (Nguyen, 2017), hence the essential part of transformed edge dynamics is:

$$\dot{\tilde{z}}_2 = (I_{N-1} \otimes A) \tilde{z}_2 + (I_{N-1} \otimes B) \tilde{w}_2$$

Equivalence between consensus and stabilization

Theorem 1

Decentralized stabilizing controller $\tilde{w} = -\mathcal{F}\tilde{z}$ for the transformed edge dynamics with $\mathcal{F} = F \otimes K$, $K \in \mathbb{R}^{m \times n}$ and

$$F = \begin{bmatrix} 0 & 0 \\ 0 & \mu\Gamma \end{bmatrix},$$

is equivalent to the consensus controller

$$u = -\mu(L \otimes K)x,$$

for the initial MAS.

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- * Stabilization here means closed-loop system $\dot{\tilde{z}} = [\bar{\Gamma} \otimes A - (I_M \otimes B)\mathcal{F}] \tilde{z}$ is stable, i.e. $\lim_{t \rightarrow \infty} \tilde{z}(t) = 0$.

Decentralized stabilizing controller design

Theorem 2

For any $\mu > 0$, the transformed edge dynamics is stabilized by the locally optimal decentralized controller $\tilde{w} = -\mathcal{F}\tilde{z}$ with the controller gain $K = R_1 B^T P_1$ and P_1 is the solution of local Riccati equation

$$P_1 A + A^T P_1 + Q_1 - P_1 B R_1 B^T P_1 = 0.$$

Accordingly, the initial MAS reaches consensus with the controller $u = -\mu(L \otimes K)x$ for any $\mu > 0$.

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- * Conditions: (A, B) is stabilizable; $Q \succeq 0$, $(Q^{1/2}, A)$ is detectable; $R \succ 0$.
- * Above approach is based on the so-called linear quadratic regulator (LQR) method.